Coventry University (CU) | National Institute of Business Management (NIBM)

DATA WAREHOUSING AND BUSINESS INTELLIGENCE

Data Concepts

BSc (Hons) Computing - Level 4 (3 Credits)

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Day 1
Foundations

Data Concepts

What data is, how it is classified, and why quality matters - before we build anything.

Session Objectives

What we are going to learn today

- **Define and differentiate between data, information, and knowledge.**
- **Recognise different types and dimensions of data.**
- **Understand the concepts of big data and metadata.**
- **Learn the basics of data storage, processing, ETL, and data mining.**
- **Recognise the need for data management for efficient data retrieval.**

A note on LO 4:

ETL and data mining appear in objective 4. We plant the seed today. Both are built in full on Days 3, 4, and beyond.

Today's focus:

LO1 - understand what data is and why we need a structured approach to managing it.

What is Data?

- **Data: raw facts and figures - unprocessed, uncontextualised, uninterpreted.**
- **Student enrollment records at NIBM: student_id, course_id, enrollment_date, grade.**
- **Colombo daily weather readings: temperature at 09:00, humidity percentage, rainfall mm.**
- **Sales transactions at Cargills Food City: item code, quantity, price, timestamp.**

What makes data 'raw'? Raw means unprocessed, uncontextualised, and uninterpreted. The number 47 alone tells you nothing. 47 what? From where? Compared to what? Context is what turns raw data into something useful.

Activity

What data did you produce today?

2 minutes - individually:
Write down three types of data you produced before arriving here today.

Think about:
Your phone • Your commute • Your morning routine

Then share with the person next to you.

Quantitative vs Qualitative Data

Think about your best friend...

Credit: careerfoundry.com

Quantitative (measurable)	Qualitative (descriptive)
• 5 feet 7 inches tall • 63 kilograms • 1 older sibling, 2 younger • Owns 2 cats • Lives ~20 km away	• Curly hair, black eyes • Great sense of humour, loud, fantastic listener • Can be impatient and impulsive • Friendly face, contagious laugh

NIBM example: Number of students who passed the exam = quantitative. A lecturer's written feedback on your essay = qualitative.
Architectural relevance: Quantitative data drives our fact table measures; qualitative data describes our dimension attributes. The distinction is foundational.

Data, Information, and Knowledge

- **Data → Information: raw facts processed to provide meaning.**
- **Information → Knowledge: understanding gained from information; the ability to act.**
- **Example: '47' is data. '47 students passed Semester 1 Computing' is information. Deciding to open an extra tutorial group based on that figure is knowledge applied.**

Analogy - The Kitchen

Familiar concept: Cooking a meal from scratch.

Structural mapping: Ingredients = data (alone, meaningless). A recipe = information (quantities, sequence, method give context). A skilled cook who adapts without measuring = knowledge applied.

Where it breaks: A recipe is static; warehouse information is queried on demand and changes as data is loaded. Bad information leads to real financial consequences - not just an unpleasant meal.

DW reconnect: Transaction records = data. A semester enrollment report = information. The registrar opening a new intake based on that report = knowledge applied. The warehouse is the kitchen that makes this possible at scale.

Discussion

Data, Information, and Knowledge in Everyday Life

5 minutes. Think of your own example - one from each level:

Data: Locations on a map.
Information: The directions and routes between those locations.
Knowledge: The traveller's insight on the best time to travel, local customs, and hidden gems.

Then discuss: Where do most organisations struggle in this chain - and why?

Types of Data

- **Structured: organised, consistently formatted, easily queried - e.g. SQL tables, spreadsheets.**
- **Unstructured: no predefined format - e.g. images, video files, social media posts.**
- **Semi-structured: some organisation, but applied inconsistently - e.g. JSON, email, PDF invoices.**
- **Cargills Food City: POS sales records = structured. Customer WhatsApp complaints = unstructured. Supplier PDF invoices by email = semi-structured.**

Analogy - The Kitchen Cupboard

Familiar concept: Organising a kitchen.

Structural mapping: Labelled Tupperware = structured. A container labelled 'rice' with mixed contents = semi-structured. An unlabelled mystery bag at the back of the freezer = unstructured.

Where it breaks: Tupperware containers are independent; database tables JOIN - linked by relationships, not isolated.

DW reconnect: A warehouse is built for structured data - the labelled Tupperware. ETL handles conversion of mixed-format sources before loading.

Discussion

The Difference - Spreadsheets vs Databases

Your assignment marks fit in a spreadsheet.
NIBM enrollment records for 50,000 students across ten years do not.

What breaks first? Think about:
Query performance • Concurrent users • Data integrity • Backup & recovery

This scale problem is precisely why data warehouses exist. Hold that thought.

Dimensions of Data

Time	Subject	Location
Historical vs real-time data. When did it happen? Over which period?	Different categories of business activity - sales, inventory, enrollment, finance.	Geographic context - regional sales figures, campus-specific enrollment data.

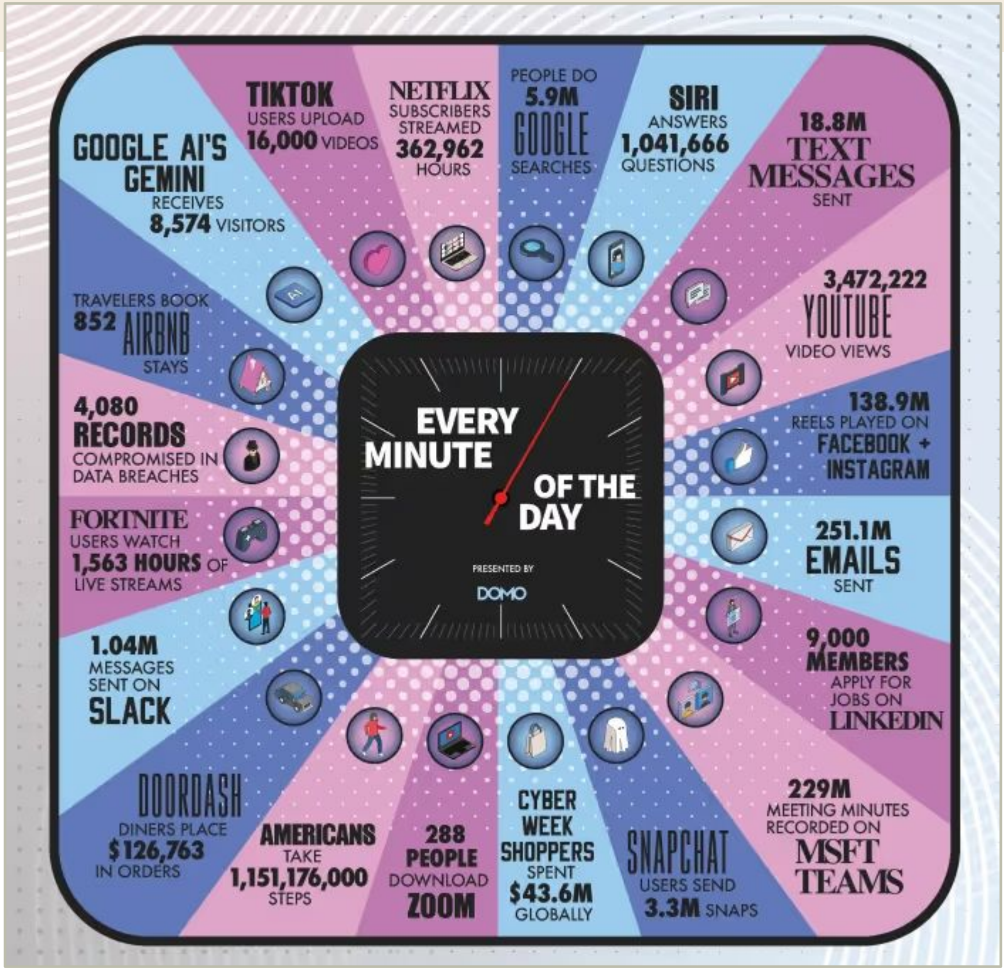
These become your dimension tables. Enrollment data has a time dimension (which semester), a subject dimension (which course), and a location dimension (which campus or delivery mode). These are literally the axes along which we slice and analyse. On Day 2, they become **dim_date, dim_course, dim_student** in our star schema.

An Internet Minute [2025]

Every minute of every day. Where does all of this go? Who analyses it? This is the problem space our module addresses.

Traditional spreadsheets and databases were not designed for this scale. This is why we need a data warehouse.

The average person produces 102 MB of data per day. At this scale, manual analysis is impossible. Architecture is not optional.



Credit: Domo | en.wikipedia.org/wiki/Internet_minute

Big Data

- Big data refers to data sets too large or complex to manage with traditional tools.
- Challenges: storage at scale, processing speed, and deriving actionable insights.
- Example: analysing social media trends during an election - millions of posts per hour.
- The 5th V - Value: big data only matters if we extract business insight from it. That is Business Intelligence - the second half of this module's title.

Analogy - The Ocean and the Harbour

Familiar concept: Navigating a vast ocean by ship.

Structural mapping: The ocean = global data. A rowing boat = traditional tools (fine on a pond, not the ocean). A cargo ship with GPS and radar = big data infrastructure. A port with warehouses = a data warehouse.

Where it breaks: An ocean is uniform (water). Big data is heterogeneous - structured, noisy, fast, slow - all at once.

DW reconnect: The warehouse is the harbourmaster's facility. It takes the cargo that matters, verifies it, and makes it accessible - via ETL.

The 5Vs of Big Data

V	What it means	Real example
Volume	Scale beyond what a single machine can hold or query quickly.	*Facebook processes over 100 billion events per day.*
Velocity	Speed at which new data arrives and decisions must be made.	*Stock exchange price ticks arriving milliseconds apart.*
Variety	Range of data types, formats, and sources arriving together.	*Video, text posts, and location check-ins from one single app.*
Veracity	Trustworthiness and accuracy of the incoming data.	*A patient allergy field left blank vs filled 'none' - both appear complete; only one is safe to act on.*
Value	The business insight or decision that justifies the entire effort.	*Netflix recommending a show you actually watch.*

The 5Vs in Practice

The same concept - five pressures operating simultaneously - appears in settings we all recognise. A hospital emergency department is a useful frame.

Analogy - Hospital Emergency Department

Volume: Hundreds of patients daily; millions of records over decades.

Velocity: Triage decisions must be made in minutes - real-time data required.

Variety: X-rays, blood test results, doctor notes, billing records - four data types simultaneously.

Veracity: An inaccurate medication record can be fatal. Data quality is a life-or-death matter here.

Value: Better patient outcomes and hospital policy - complexity only justifies itself through that.

Where it breaks: Hospitals handle individual records; big data works at aggregate scale. Also: nightly batch ≠ real-time triage.

DW reconnect: A hospital warehouse answers 'What is our 30-day readmission rate by age group?' - operational chaos becomes population-level pattern.

Data Quality Dimensions (DAMA UK) - Part 1

1. Accuracy How well data represents real-world facts or events.

Example: *A delivery service ensuring a customer's address is correctly recorded.*

2. Completeness The proportion of stored data vs what should be available.

Example: *A healthcare database with all necessary patient records and medical history details.*

3. Uniqueness Whether data is stored without unnecessary duplication.

Example: *A national ID number that appears exactly once in government databases.*

Data Quality Dimensions (DAMA UK) - Part 2

4. Consistency No conflict between data from multiple sources.

Example: *Financial reports from different branches must not show conflicting revenue figures.*

5. Timeliness Data is up-to-date and available when required.

Example: *Real-time updates on public transportation schedules for commuter reliability.*

6. Validity Data conforms to defined business rules or formats.

Example: *Customer phone numbers must follow a valid format in a telecommunications provider's database.*

These six dimensions are what ETL enforces before data enters a warehouse. A violation means data is either rejected or flagged for review. We build these checks on Day 3.

Metadata - Data About Data

- **Metadata: data that describes other data - provides context and meaning.**
- **Technical metadata: schema definitions, data types, table relationships.**
- **Business metadata: what 'revenue' means in this specific organisation - not just a number.**
- **Operational metadata: when was this data last loaded, and by which ETL process.**

Analogy - The Library Catalogue

Familiar concept: A library catalogue card - physical or digital.

Structural mapping: The card describes the book without opening it: title, author, subject, shelf location, publication date. In a warehouse, metadata tells you which table holds which records, what each column means in business terms, when data was last refreshed, and who is authorised to query it.

Where it breaks: A catalogue card is written once and rarely changes. Warehouse metadata must evolve continuously - new tables, new source systems, changing business definitions. Metadata maintenance is active and ongoing, not a one-time task.

DW reconnect: Without metadata, a warehouse is a building with no labels on any shelf. The data exists, but no analyst can locate or trust what they retrieve. Metadata is not an afterthought - it is the foundation of a usable warehouse.

Discussion

Metadata in a University Setting

3 minutes. Think about your student record at NIBM.

What metadata describes it? Consider:
- When was the record created?
- Which department owns it?
- Who is authorised to view it?
- What format is your student ID?
- How long must the record be retained by law?
- Where is it physically stored?

Share examples.

Data Storage and Processing

Storage Options

- Local storage: fastest access, limited scale, single point of failure.
- Cloud-based storage: scalable on demand, accessible from anywhere, pay-as-you-go.
- Distributed storage: data spread across many nodes - redundant, fault-tolerant, designed for big data scale (e.g. HDFS).

Processing: Batch vs Real-Time

Batch processing

Large data sets processed at a scheduled time. NIBM generating semester-end grade reports overnight - acceptable to wait until morning.

Real-time processing

Data processed immediately as it arrives. A bank detecting a fraudulent card transaction as it happens - delay is not acceptable.

Introduction to Data Management

- **Data management: organising and maintaining data assets to meet information needs across an organisation.**
- **Ensures data quality, security, and accessibility throughout the data's lifecycle.**
- **Covers governance, stewardship, architecture, and storage - not just databases.**
- **What happens without it? Silent data corruption. Conflicting reports from different departments. Compliance failures.**

Analogy - Library
Data management is like maintaining a well-run library:

• Orderly - everything in its place.
• Secure - some sections restricted.
• Accessible - catalogue makes finding easy.
• Maintained - new books added, old ones retired.

Without the librarian, the books still exist - but finding anything becomes impossible.

Facilitating Data Retrieval

Key requirements for efficient retrieval

- Organised storage: data must be structured consistently before retrieval is possible.
- Efficient indexing: the query engine needs indexes to avoid full-table scans on millions of rows.
- Metadata use: catalogue tells the system where to look before it looks.

Challenges in practice

- Large volumes: queries that touch millions of rows must be designed carefully - full scans are expensive.
- Varied formats: joining structured and semi-structured data requires transformation first.
- Data accuracy: the system retrieves exactly what was stored - garbage in, garbage out.

KEY CONCEPT

From Data Concepts to Data Warehouse

Everything - **what data is, how it is classified, why quality matters, what metadata does -** is not background theory. It is the foundation that makes the architecture of the next six days coherent.

You cannot build a reliable warehouse on data you do not understand. Tomorrow we begin designing the schema. From Day 1's OLTP tables to Day 2's star schema, the decisions we make will be grounded in what you have learned today.

Questions & Answers

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Next: Foundations (DSS Evolution, OLTP vs OLAP, Warehouse vs Database)

Data Concepts

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